Documentary Evidence Report

Data lineage of exp1 and exp2 and overlap between training and field data

Project: Helipad Detector. Prepared on: October 14, 2026. Basis: materials from the project proponent's repository, Pedro Vyctor Almeida's repository, Roboflow screenshots, and the helipad_dataset.rar archive, without relying on contact with third parties or filling gaps through assumptions.

Responsible-use notice. Automated image-based visual detection is subject to errors and is not equivalent to operational, legal, safety, or aeronautical validation. This report addresses data lineage, not site validation.

1. Conclusions

Each conclusion states its level of support: proof (verified in files), circumstantial evidence (consistent with multiple elements, but without a direct record), or unconfirmed.

- **exp1 (strong circumstantial evidence).** The exp1 weights and results in the proponent's repository are Pedro Vyctor Almeida's "exp1-2" run (proof: byte-for-byte identical files), configured for the My-First-Project-2 folder, i.e., dataset version 2 (178 training, 34 validation, 14 test images). The log showing 116/33 images saved in the notebook is consistent with another run, the original one (version 1), for which only args.yaml remains. The image-count log for "exp1-2" is missing; therefore, the link to version 2 is strong circumstantial evidence, not proof.
- **exp2 (strong circumstantial evidence, with a conflict to resolve).** The preserved log (178 training, 34 validation, 0 corrupt images), results.csv, and best.pt are mutually consistent and correspond to the contents of Pedro's version 2, not to the fork versions shown in the screenshots (356 training images with data augmentation, 404 in total). According to the log, the fork-based run is exp3. The claim that exp2 used the fork is not supported by the files; see Section 4.
- **exp1 × exp2 comparison.** Both point to the same data contents (the archived copies of the two datasets are identical), but different people executed them, in different environments and Ultralytics versions (8.4.75 and 8.4.98), and with different numbers of epochs (60 and 100). This is therefore not a controlled experiment; data equivalence is supported by circumstantial evidence, not proof.
- **Field data.** The set of 7,767 region × tile pairs was reconstructed from the repository's compressed archives and matches the historical count, but contains only 5,540 distinct tiles (neighboring regions overlap).
- **Overlap.** Of the 225 annotated images with coordinates, 125 (87 of 177 training, 25 of 34 validation, 13 of 14 test images) are copies of a field tile (proof through pixel comparison). These 125 tiles represent 2.3% of the 5,540 distinct field tiles. There are no direct copies of validation or test images in the training set; spatial proximity (within 1 tile) occurs for 15 of the 34 validation images and 3 of the 14 test images.
- **Filenames are not sufficient.** Twenty-three images (17 training, 4 validation, 2 test) share the same tile identifier in their filenames (x194146_y297518_itaim_bibi-NN-); the contents of 22 of them match other field tiles. Analysis based on coordinates read from filenames would overestimate or mislocate the overlap; this is why the proof is pixel-based.

2. Revisions and method

Table 1. Revisions and sources used

Source	Pinned revision	Use
Proponent's repository (Mindful-AI-Research/3-project-ai-ml-yolo-helipad_detector)	bc5a0ad953e4ffc8433d3e3f68d245068ebddc5a (October 13, 2026, 15:20 -03): datasets archived in data/training/exp1 and exp2, data/raw/helipad_dataset.rar, artifacts in artifacts/runs/runs/detect/exp1–exp4	Archived data and artifacts
Same repository, main branch	9e767ec1f8d711b4e09f920aadf802c3855acb9a (October 14, 2026, 02:14 -03): notebooks src/training/yolo_training_exp1–4.ipynb, src/geospatial/*, field reports	Saved code, reports, and tile ZIP archives
Pedro Vyctor Almeida's repository (ppvyctor/Helipoint-Detector)	d852e3c73dc9f65a151349d8883a0acc484c0f97 (June 24, 2026, 20:11 -03); runs/detect/runs/exp1-2 and My-First-Project-2 present since bcefd53 (June 22, 2026, 20:46 -03)	Copy of the exp1-2 run and data folder
Roboflow screenshots	Four screenshots dated October 14, 2026 (Pedro's version 2; attempt to open version 1; fork versions and model; fork v2 details)	Version counts and dates

Only files and screenshots were used; no API keys are reproduced (the saved notebooks read them from environment variables or Colab Secrets). The .pt checkpoints were inspected without executing code (reading data.pkl with pickletools); labels.cache files were inspected using a restricted deserializer. The current notebook code describes what is saved today, not what was executed; the conclusions therefore rely on output artifacts (args.yaml, results.csv, weights, preserved logs), and folder names alone were not treated as proof.

3. Comparison between exp1 and exp2

Table 2. Provenance of each data item for exp1 and exp2 (exp3 and exp4 as context)

Item	exp1	exp2	Source and level of support
Dataset origin	Project my-first-project-kfo4l, workspace pedros-workspace-1w4rz (Pedro)	Same project (Pedro); notebook with an assert against the fork	README.roboflow in the archived datasets; exp2 notebook, cell 38. Proof (archived contents); code is circumstantial evidence of use
Run authorship	Pedro (weights identical to those in runs/detect/runs/exp1-2 in his repository)	Proponent (self-reported); Colab T4, Ultralytics 8.4.98, July 17, 2026	SHA-256 of weights (proof); exp2 authorship is the proponent's statement, consistent with date and environment
Roboflow version	v2 (circumstantial evidence: args.yaml points to My-First-Project-2)	v2 (log: 178/34 in My-First-Project-2)	`args.yaml; exp2 training cell. Strong circumstantial evidence (exp1 and exp2)`
Archived dataset	`data/training/exp1: 226 images (178/34/14), no augmentation, exported June 22, 2026, 22:34 GMT`	`data/training/exp2: identical to exp1`	README.roboflow and file comparison. Proof that the two copies are identical, not that they were used for training
Current saved code	exp1 notebook points to My-First-Project-1, 60 epochs; log 116/33	exp2 notebook: Pedro/v2 with assert; 100 epochs	Notebooks on main. The exp1 notebook does not correspond to the preserved metrics
Preserved output	Log with 116/33 (My-First-Project-1), consistent with args.yaml from Pedro's original exp1 (v1, June 21); no image-count line for exp1-2	Log with 178 training images, 1 background, 0 corrupt; 34 validation images	Notebooks. exp1: missing log; exp2: proof of the contents counted
Metrics	`results.csv: 60 epochs; best fitness at epoch 54: P 1.000, R 0.963, mAP@50 0.994, mAP@50-95 0.881`	`results.csv: 100 epochs; best at epoch 81: P 0.982, R 0.971, mAP@50 0.993, mAP@50-95 0.888`	`results.csv. Proof: values match the table in the project report`
Weights	`best.pt saved June 22, 2026, 22:42:39; Ultralytics 8.4.75; SHA-256 0f8ce4ed…, identical to Pedro's`	`best.pt saved July 17, 2026, 03:41:53 (time zone not recorded); embedded metrics P 0.98186, R 0.97059, mAP@50 0.99338, mAP@50-95 0.88759`	Checkpoint inspection. Proof: embedded metrics = epoch 81 of results.csv
Dates	Version generated June 22, 19:29 (screenshot time); exported 22:34 GMT; weights 22:42 (time zone not recorded); Pedro's commit June 22, 20:46 (-03); commit in the proponent's copy September 27, 2026	Weights July 17; archived datasets committed July 17, 14:08 (-03); artifacts on September 27, 2026	Screenshots, README, and git log. Sequence consistent with version 2; commit dates do not prove contents
Context: exp3 and exp4	exp3: fork, My-First-Project-1, 356/34 (augmentation), 8.4.92, July 17, 04:27	exp4: Pedro v2, 178/34, 8.4.163, September 26, YOLO11n; 3 labels flagged	Notebook logs (proof of counts)

3.1 Conclusion on exp1

Table 3. Level of support, exp1

Level of	Claim and basis
Proof	(a) The exp1 best.pt and last.pt weights in the proponent's copy have the same SHA-256 as those in runs/detect/runs/exp1-2 in Pedro's repository. (b) The args.yaml for these weights records data: My-First-Project-2/data.yaml, name: exp1-2, 60 epochs, seed 42. (c) The exp1 rows in the project report's table match results.csv for this run (epoch 54). (d) Pedro's repository also retains args.yaml from an earlier exp1 (data: My-First-Project-1/data.yaml, 60 epochs, seed 42; commit dated June 21, 2026), without results or weights. The notebook log showing 116/33 images (My-First-Project-1) is consistent with that original run, not v2.
Strong circumstantial evidence	The My-First-Project-2 folder follows the SDK's naming convention (project name and version). The fork versions shown in the screenshots are dated July 10, 2026, after the exp1-2 weights (June 22); this folder is therefore very likely Pedro's v2. Version 2 was generated at 19:29 (screenshot time) and exported at 22:34 GMT; the weights are timestamped 22:42 on the same day; Pedro committed them to his repository at 20:46 (-03), approximately one hour later (assumed time zones: screenshot -03, checkpoint UTC).
Unconfirmed	Whether exp1-2 counted 178/34 images (no preserved Scanning line); results and weights from the original exp1 (v1), which are not in the materials; the contents of Pedro's v1 (the /dataset/1 page opened blank, which does not prove nonexistence).

Effect on the preliminary research proposal. Rather than being treated as having "inconsistent lineage," exp1 is now recorded as version 2 on the basis of documentary evidence, subject to the caveat above.

4. exp2, fork, log, metrics, and weights

The proponent states that she executed exp2 and used a fork of Pedro's dataset. Separating dataset origin from run authorship: the origin is Pedro (project my-first-project-kfo4l); the run was executed by the proponent, which is consistent with the date (July 17), Colab with a T4, and Ultralytics 8.4.98, which differ from exp1, but is not proven by a file. The fork-related claim conflicts with the artifacts:

Table 4. Fork, log, metrics, and weights

Link	Evidence	Interpretation
Fork (screenshots)	`fabiana-campanari-workspace/my-first-project-kfo4l-jcbdq: versions v1 and v2, both with 404 images; v2: 356 training, 34 validation, 14 test, stretch 640×640, augmentation (2 outputs per example, flips, ±15°, brightness ±25%)`	The total and split are not those of exp2 (178/34/14, no augmentation). Details of v1 contents are not shown in the screenshots
exp2 log	178 training images, 34 validation images, 0 corrupt, in /content/My-First-Project-2	The SDK names the folder using the project and version, and the fork is also called "My First Project": the name does not distinguish the origin, but the count does. 178 is consistent with Pedro's v2 and incompatible with the fork v2 shown (356)
exp3 log	356 training, 34 validation, My-First-Project-1; code downloads version 1 of the fork	The fork-based run is exp3. The project report also describes the change: the first fork-based run was saved as "exp2", renamed exp3, and exp2 was rerun using the original dataset
`exp2 results.csv and best.pt`	100 epochs; best epoch 81; metrics embedded in best.pt = epoch 81 row	Mutually consistent (proof). Linked to the log through args.yaml: data: /content/My-First-Project-2/data.yaml, name: exp2
Roboflow dashboard model (fork v2)	`my-first-project-kfo4l-jcbdq-2-yolov8s-t1, Roboflow 3.0 Object Detection, updated July 10, 2026, 2:40; test: mAP@50 92.1%, precision 100%, recall 92.9%, F1 96.3%`	A different training run ("s" architecture, Roboflow platform, fork dataset, July 10). It is not the exp2 notebook run, and its metrics must not be attributed to it

Conclusion. The four exp2 artifacts (log, results.csv, best.pt, args.yaml) form a coherent set and link to the contents of Pedro's v2. To reconcile this with the proponent's statement, two possibilities remain that the files do not distinguish: (i) the current exp2 is the rerun with the original dataset, and the fork-based run is exp3; (ii) a third fork version exists, without augmentation (178/34), which is not shown in the screenshots. Possibility (i) is supported by the project report and logs; (ii) is unconfirmed. How the 178/34 version was obtained for exp2 remains to be clarified.

Caution. Even with equivalent data, exp1 × exp2 is not a controlled comparison: the person executing the run, environment, Ultralytics version, and number of epochs vary. The same applies to exp2 × exp4 (8.4.98 and 8.4.163). The "3 corrupt" flag in exp4 has a verifiable cause in the data: three training label files (tile_z19_x194014_y297336, x194018_y297341, x194026_y297448) mix polygon and bounding-box rows, and the log states exactly "labels mix segment and detection rows". It was not possible to rerun earlier Ultralytics versions to determine whether they handled these three files differently. Overall, 170 of the 178 training label files contain only polygons (as do all validation and test label files), although the documentation refers to bounding boxes.

5. Field data: tile inventory

The historical 7,767 region × tile pairs were reconstructed from the repository's mosaic_* ZIP archives (8,083 raw entries): the inventory (field_tiles_inventory.csv, containing the original name, normalized identifier, region, z/x/y, path within the ZIP, CRC32, and number of regions containing the tile) has exactly 7,767 rows, and the 7,943 reported in the production report are reproduced from the bounding boxes in sp_neighborhoods_bbox.csv.

Table 5. Field inventory totals

Measure	Value
Raw entries in ZIP archives	8,083
Region × tile pairs (inventory rows)	7,767 (same as historical record)
Distinct tile identifiers	5,540
Identifiers present in more than one region	1,958 (identical contents, same CRC32)
Production grid reproduced from bounding boxes	7,943

Gaps: for Faria Lima, the historical record indicates 840 tiles and the ZIP archives contain 664 distinct tiles (176 missing, not identified; the ZIP archives contain 980 entries, including duplicate parts); splitting Av. Paulista into two sections shifts 5 tiles between the production grid and the set of 7,767.

6. Overlap between training and field data

Criteria. (i) Exact pixel copy: mean absolute error (0–255) less than or equal to 5 between the annotated image and the most similar field tile (32×32 prefilter selecting the top three candidates, measured at 256×256). All matches are at 1.56 or below; no other candidate is below 16.9, so the result is not sensitive to the threshold. (ii) Filename-based identifier: z/x/y read from the filename, excluding Roboflow suffixes; used only when unique (the 23 filenames sharing the same coordinates were resolved through pixels). (iii) Spatial proximity: Chebyshev distance, in z19 tiles (approximately 70 m), to the nearest field tile, with k = 1, 2, 3, and 5; proximity is not confirmed leakage. Denominators: 177, 34, and 14 images with coordinates (plus mosaico_Taquara_hires_preview, which has no tile ID and no match, bringing training to 178); 5,540 distinct field tiles; 7,767 region × tile pairs.

Table 6. Annotated images and field tiles

Measure	Training (177)	Validation (34)	Test (14)	Total (225)
Exact copy of a field tile (pixels)	87 (49.2%)	25 (73.5%)	13 (92.9%)	125 (55.6%)
No copy, but within ≤ 1 tile of field data	0	0	0	0
No copy, within ≤ 2 / 3 / 5 tiles	2 / 3 / 8	1 / 2 / 2	0 / 0 / 0	3 / 5 / 10
No pixel match	90	9	1	100
With a shared tile name	17	4	2	23

The 125 copied tiles are distinct from one another and correspond to 212 region × tile pairs (2.73% of 7,767) and 2.26% of the 5,540 distinct tiles. They are concentrated in Av. Paulista (33 tiles across both sections), Brooklin (20, exclusive to the region), Inter Zonas and Vila Olímpia (18 shared between the two), and Alphaville Industrial (15); the full list is in the CSV's field_regions column. The 100 unmatched images are outside the bounding boxes of all regions; according to their coordinates, approximately 70 fall within an approximate rectangle of the municipality of São Paulo and 30 are elsewhere in Brazil (91 training, 9 validation, 1 test). This rectangle is a rough approximation used only for descriptive purposes.

Training × validation and test. Pixel comparison shows that no validation or test image is a copy of a training image (minimum error of 31.6 for validation and 42.7 for test), and there are no copies among training images. The earlier filename-based finding of 4 validation images and 2 test images sharing a tile with training was an artifact of the shared identifier. Spatial proximity does occur: 15 of the 34 validation images and 3 of the 14 test images are within 1 tile of a training image (20 and 11 within 2 tiles), reducing the spatial independence of validation.

Interpretation. Training already contains 87 tiles that are part of the field data (1.6% of distinct tiles), which may inflate the detection rate on those tiles; the effect on overall rates cannot be calculated without per-tile detections for each experiment. The 22 cases in which image contents differ from the filename (suffix itaim_bibi-NN-, all sharing the same coordinates) match distinct tiles in the Itaim Bibi region by pixel comparison (error of 0.2 to 0.3 at 32×32); the cause of the repeated filename (renaming, export) is unknown from the supplied materials.

7. Limitations

- Cropped, rotated, mirrored, rescaled copies, or copies from another date, are not detected by the pixel criterion; actual overlap may be greater than measured.
- The materials contain no per-tile detections; no effect on field rates was calculated, and no inference was rerun.
- The prefilter uses the top three candidates at 32×32; matches ≤ 1.56 and the gap up to 16.9 indicate that the criterion is stable, but this is not an exhaustive search at 256×256.
- The contents of data/raw/helipad_dataset.rar were opened: 174 tile files (121 distinct identifiers, all in the field inventory), and 110 annotated images are copies of files in that archive. How the archive was assembled is unknown (unknown from the supplied materials).
- The logs for the original exp1, Pedro's v1, and the fork-based run before exp3 were not located; the blank /dataset/1 page does not prove nonexistence.
- Counts, not individual tiles, were identified for Faria Lima (176 missing) and Av. Paulista (5).

8. Files still to be attached

Table 7. Outstanding items needed to close the gaps

File or information	Gap addressed	Decision affected
Log of the exp1-2 run with the Scanning lines, or training output from that run; results and weights from the original exp1 (v1), if they exist	Image count for the exp1 run and what happened to the original run	Upgrade exp1 from strong circumstantial evidence to proof
Export or screenshot of the fork's versions page showing the split for each version (v1 and v2) and dates, plus the list of projects in the proponent's workspace	Whether a fork version with 178/34 exists and which version was used in each run	Resolve the exp2 conflict
Artifacts (args.yaml, results.csv, weights) from the first fork-based run, before it was renamed exp3	Whether the "silent exp2" run is the current exp3 or another run	exp3 lineage
Pedro's Heliponto folder (Heliponto.rar) and information on how helipad_dataset.rar was assembled	Origin of the 100 images without a field tile	Geographic coverage and data origin
Per-tile detections (CSV per experiment) for the 7,767 pairs	Effect of overlap on field rates	QE1: share of the difference explained by lineage
List of the 176 missing Faria Lima tiles and the 5 Av. Paulista tiles	Complete the inventory (840 vs. 664)	Field denominator
Output from ultralytics 8.4.98 and 8.4.163 on v2 (label verification)	How each version handles the 3 mixed-format labels	Comparability of exp2 and exp4